

# Eightfold Candidate Transformer - Technical Design

## Pipeline

Detect source type, extract fields, normalize formats, merge candidate evidence, score confidence, project to the requested customer schema, then validate before output.

## Canonical schema

candidate\_id, full\_name, emails, phones, location, links, headline, years\_experience, skills with confidence/sources, experience, education, provenance, and overall\_confidence.

## Normalized formats

Emails are lowercased, phones are E.164, countries are ISO-3166 alpha-2, dates are YYYY-MM, and skills are canonical lowercase names using an alias map such as JS -> javascript.

## Merge policy

Records match by email first, phone second, normalized name last. Scalar conflicts are resolved by repeated evidence, average source confidence, then individual confidence. Lists are deduped.

## Runtime config

The transformer always builds a full canonical record first. A separate projection layer selects fields, renames paths, maps from canonical paths like emails[0] or skills[].name, applies optional normalizers, toggles confidence/provenance, and handles missing values as null, omit, or error.

## Handled edge cases

Missing or malformed source fields do not crash the run; bad phones are dropped instead of guessed; CSV and notes for the same person merge into one profile; every chosen value keeps provenance.

## Scoped out

ATS JSON, live GitHub/LinkedIn calls, and PDF/DOCX resume parsing are left as future extractors. They can emit the same intermediate evidence objects without changing merge or projection logic.